

ProLiVis 2.0 — Supplementary Material

A walkthrough of every capability

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1 What this document is

The manuscript argues one thing: that a protein interaction database records evidence without weighing it, and that a literature-centric view with a trust model on every edge makes the difference visible. This supplement is the other half — what the tool actually does, capability by capability, with a figure for each.

Everything shown here comes from BioGRID release 5.0.260, the coronavirus file (76,632 records, 4.5 MB), restricted to SARS-CoV-2 unless stated: a reader can download it and reproduce every figure. Where a capability only means something at scale, the manuscript reports the full release instead.

Figures and tables are generated rather than assembled by hand:

```
cd web
PROLIVIS_FIGURE_SOURCE=~/.Downloads/BIOGRID-CORONAVIRUS-5.0.260.tab3.zip \
npm run supplement
```

Interface or API. ProLiVis has two surfaces, and this document is explicit about which one each capability lives on. The interface is what a browser tab shows. The scripting API is `window.prolivis`, available in any browser console and in headless automation; it is a supported interface, not a debugging hook — the tool’s own tests and every figure in the manuscript are produced through it. Table 1 is the map. Four view models are computed but not yet drawn (§11); we would rather say so than imply otherwise.

Table 1: Where each capability lives.

Capability	Interface	Scripting API
Ingest a BioGRID file	yes	<code>load, inspect</code>
Online mode (BioGRID REST)	no	<code>connect, fetchRemote</code>
Literature (center) view	yes	<code>centerLayout, centerScene</code>
Filter the literature	yes	arguments to <code>centerLayout</code>
Protein network, five layouts	yes	<code>graph, networkLayout</code>
Modules view and drill-down	yes	<code>autoContract, highLevelLayout</code>
Adjacency matrix	yes	<code>matrix</code>
Trust threshold	yes	<code>score</code>
Trust weights and presets	no	<code>trustPresets, gather, rescore</code>
Calibration and ablation	no	<code>calibrate, ablate</code>
Graph algorithms	no	<code>cliques, modules, cores, ...</code>
Literature search and collection	yes	<code>findPublications, derive</code>
Compare and merge	yes	<code>compare, merge</code>
Export a figure as SVG	yes	<code>toSvg</code>
Export data (CSV, GraphML, SIF)	no	<code>exportTable, exportGraphml, ...</code>
Session manifest	no	<code>manifest, parseManifest</code>
UpSet, timeline, chord, bipartite	no	<code>upset, timeline, ...</code>

2 Getting data in

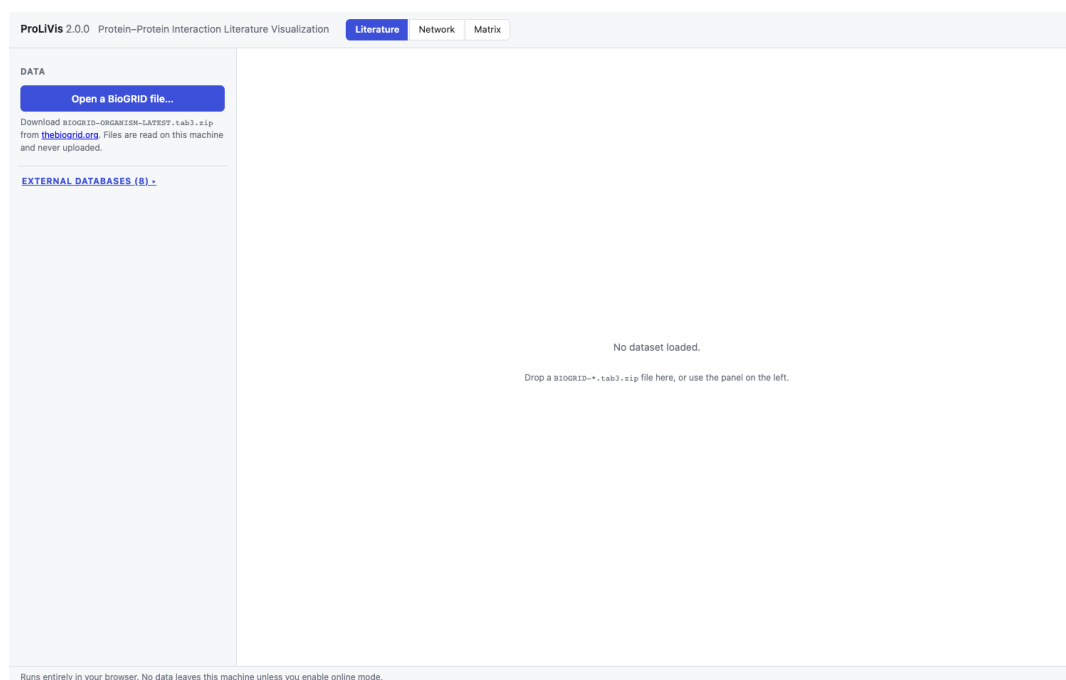


Figure 1: The application before any data. There is no account, no server and no upload: a BioGRID archive is opened from disk and parsed in the browser. The panel says where to get one.

A dropped `.tab3.zip` is inflated in line-aligned chunks with the reader throttled, so peak memory is bounded by one chunk rather than by the archive; members are listed first, and an archive holding one file per organism asks which. Columns are resolved by alias onto a canonical set, because the bulk `tab3` files and the `tab2` output of the REST service name six columns differently. The result is stored in DuckDB compiled to WebAssembly and persisted in the browser's Origin Private File System, so a second visit opens immediately; if the browser refuses persistent storage — a private window, or a second tab already holding the database — the application says so and continues in memory rather than failing.

3 The literature view

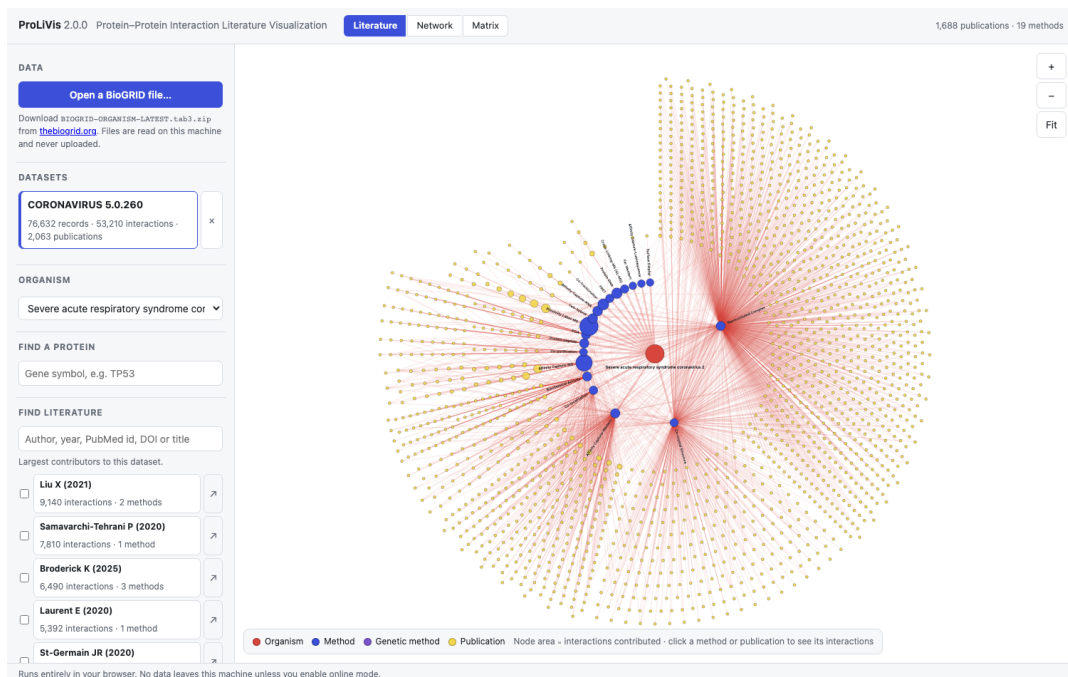


Figure 2: Center Layout 2.0 for SARS-CoV-2. The organism is at the centre, experimental methods form the inner ring, and publications fan outward within their method's sector. Sector width is share of the literature; node area is interactions contributed. Genetic methods are drawn in a different colour, because a genetic interaction is not a physical one.

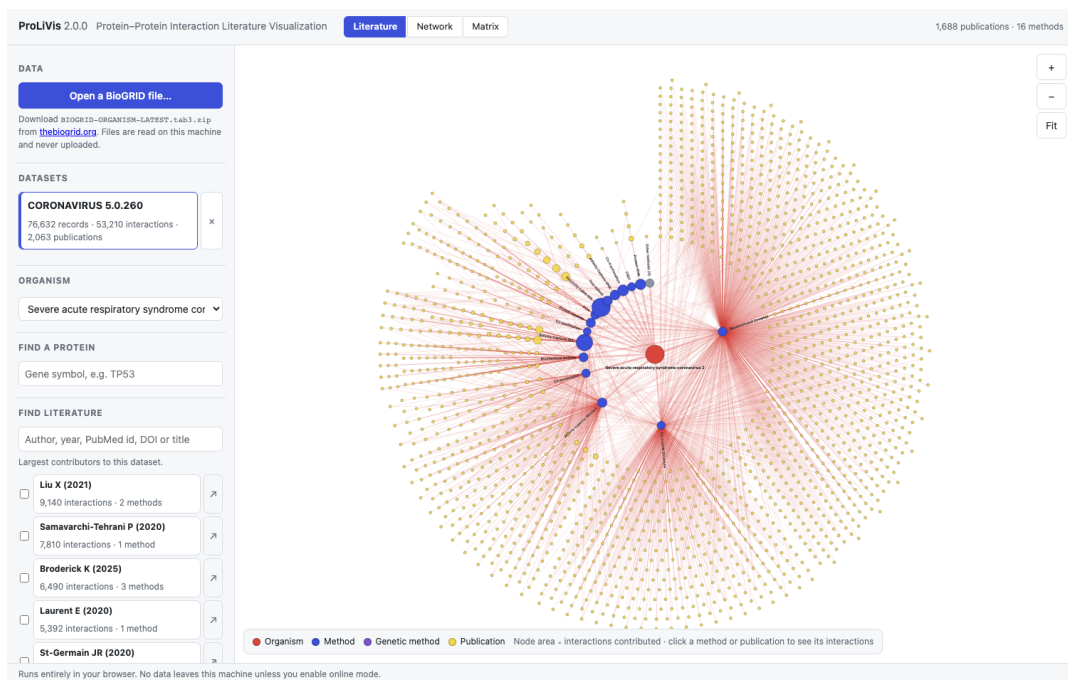


Figure 3: The same literature with methods supported by fewer than five publications folded into a single node — the paper’s third level. The long tail otherwise consumes half the circle in slivers too thin to read or click.

4 Filtering the literature

Every publication is one node whatever it reported, so a screen contributing ten thousand interactions and a structure paper contributing one look alike until you ask. Three filters ask: contribution, proteins touched, and method.

FILTER PUBLICATIONS [clear](#)

Drawing **85** of 1,688 publications. Counted within this organism and the methods selected — the same quantity the node sizes encode, so a paper's total across the whole release can be larger.

Interactions per publication

to

1–7,810 in this literature.

Proteins per publication

to

1–2,757 in this literature.

Methods all 19

☒ Reconstituted Complex 1,082

☒ Co-crystal Structure 477

☒ Affinity Capture-Western 307

☒ Co-localization 95

☒ Biochemical Activity 91

☒ Affinity Capture-MS 78

☒ Co-purification 49

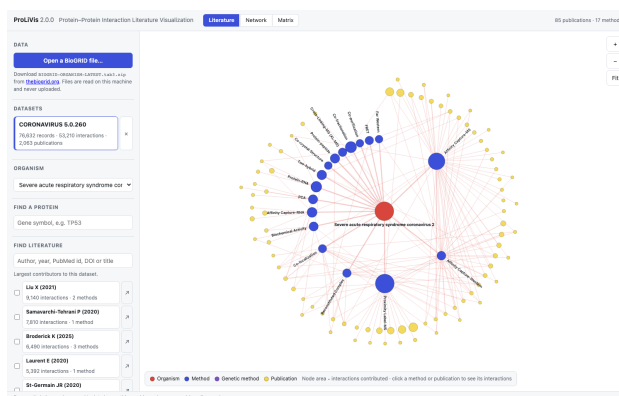


Figure 4: The filter, and its effect. Bounds beside each box are the range of the unfiltered literature, so they stay put as the view narrows. Counts are within the selected organism and methods — the same quantity the node areas encode. Filtering recomputes the method ring from the publications that survive, since a sector's width is its share of the literature and a sector sized by papers that are no longer drawn would describe a different picture.

The publication band is bounded, so a literature larger than it holds is not drawn in full; the panel reports what it drew against what exists, and what matched but did not fit. This matters at scale rather than here: the human literature of this release is 41,218 publications, of which 16,148 fit.

5 From a publication to what it reported

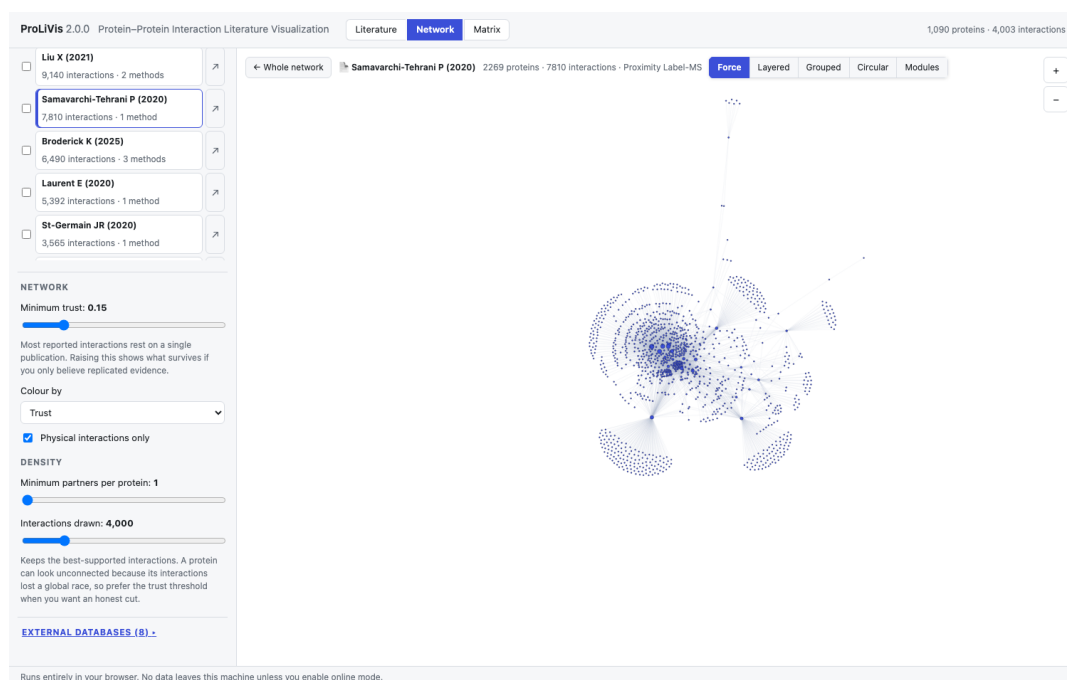


Figure 5: Clicking a publication node opens the interactions that paper reported. The breadcrumb above the canvas names the scope and returns to the whole network. A method node does the same one level up: every interaction that technique produced. Inside a scope the trust threshold does not apply — you asked for a specific bounded set — so trust shows as edge colour and weight instead of hiding anything.

6 The protein network

The figures below are the same network under each arrangement: SARS-CoV-2 at trust ≥ 0.4 , keeping proteins with at least two partners and the 400 best-supported interactions — 48 proteins and 248 interactions. That is a deliberate choice for this document. The whole organism is 8,510 proteins and 34,540 interactions, and a figure of that says only “it is large”; what an arrangement *means* is legible at fifty nodes.

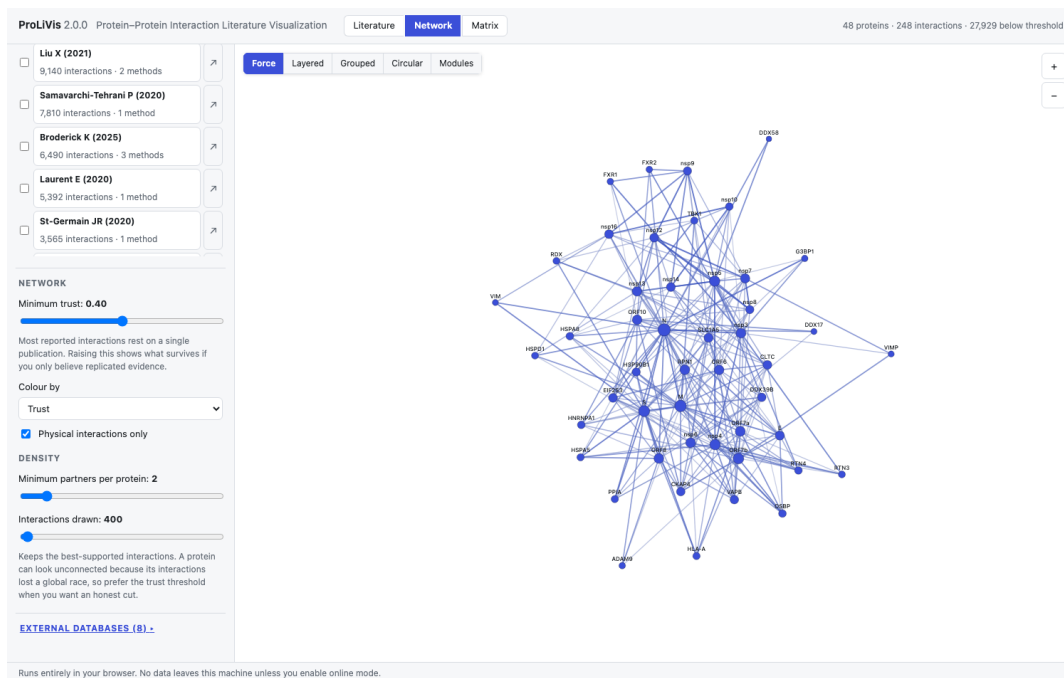


Figure 6: Force. What clusters together. The replication machinery (nsp5, nsp9, nsp10, nsp12–nsp16) sits together, with host chaperones and stress-granule proteins (HSPA5, HSPA8, G3BP1, DDX58) around it. The density controls used are on the left; the header reports how many interactions the threshold hid.

The four arrangements are all deterministic: a seeded generator, a fixed iteration count rather than a run to convergence, and locale-independent tie-breaking, so the same network always draws the same picture and a figure can be regenerated from a session manifest.

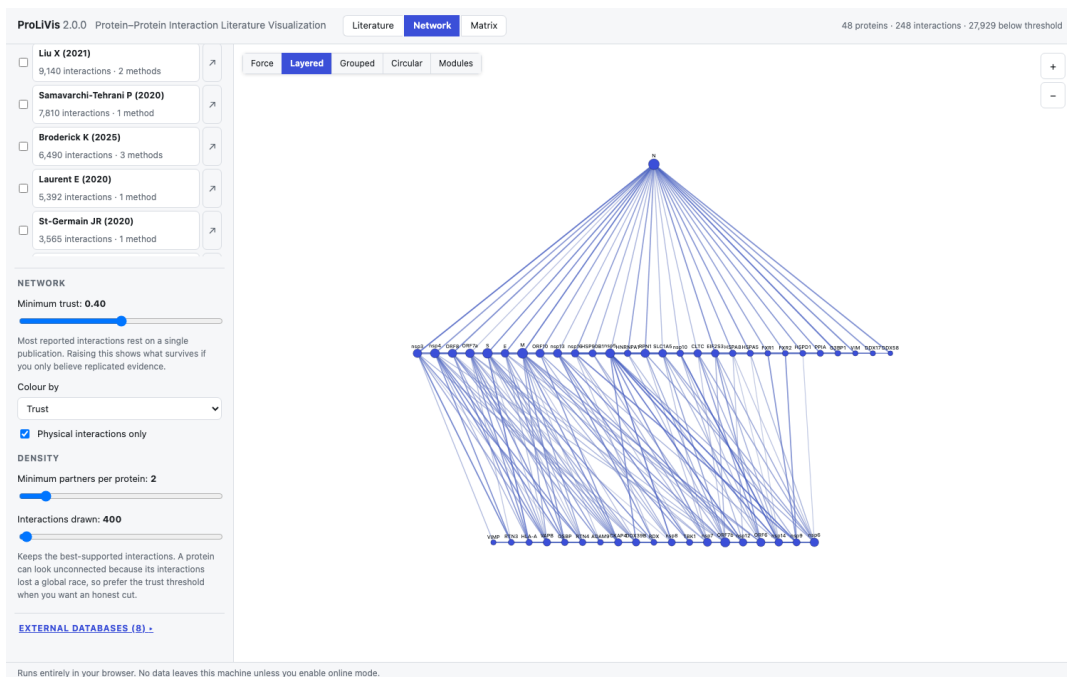


Figure 7: Layered. Vertical position is graph distance from the best-connected protein, so “how many hops apart are these” can be read off the picture. Layers wider than the canvas wrap into sub-rows rather than overflowing.

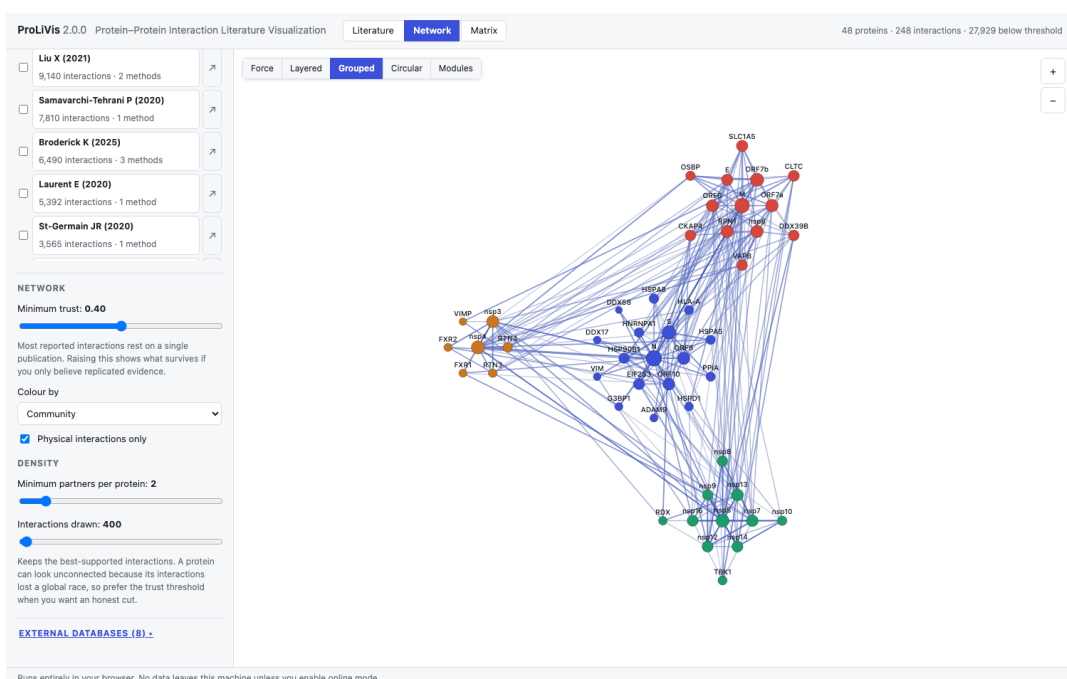


Figure 8: Grouped, coloured by community. Trust-weighted communities as discs, packed around the largest, best-connected member at the centre of each. Grouping by connected component, the obvious choice, does not work here: a PPI network is one component, so every protein lands in one group and the arrangement draws a single disc.

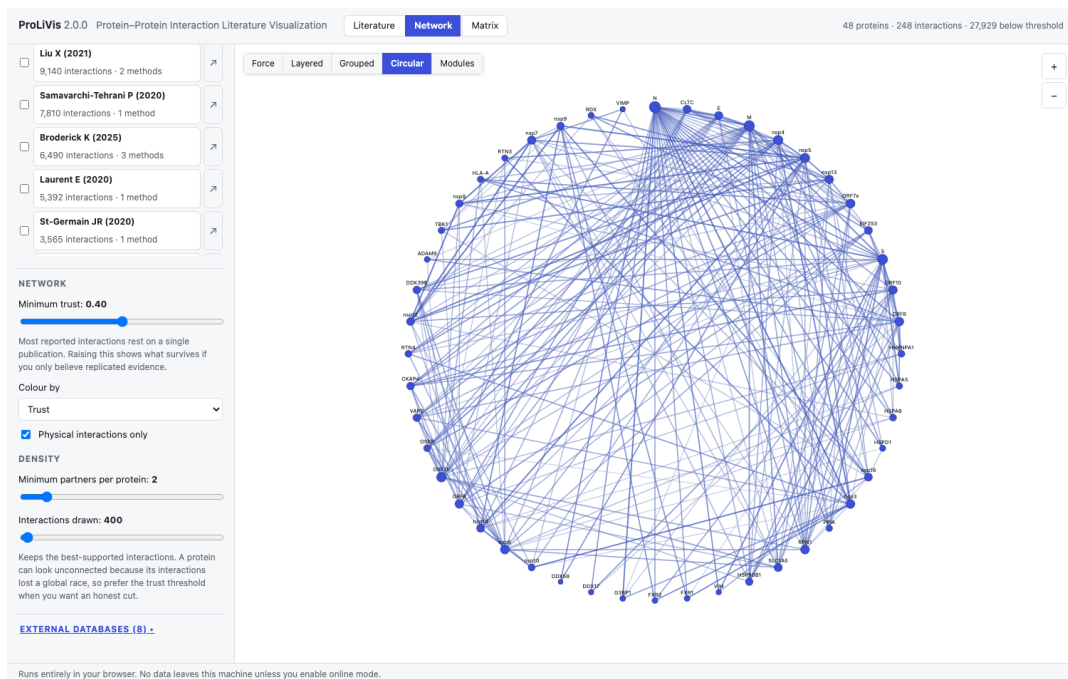


Figure 9: Circular. Everything on one ring, ordered by a traversal so neighbours sit together: a stable reference arrangement that makes no claim about structure.

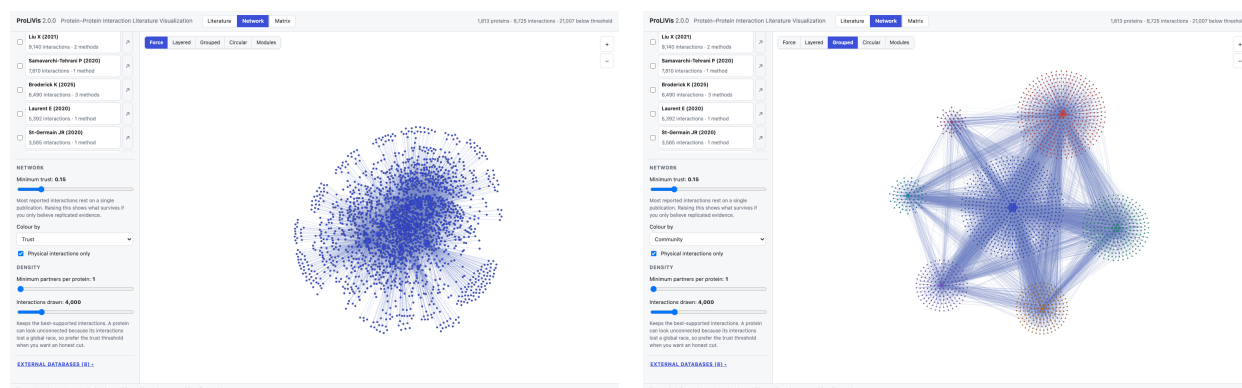


Figure 10: The whole organism at the default threshold — 1,813 proteins and 8,725 interactions — under Force (left) and Grouped (right). Force uses Barnes–Hut repulsion above three hundred proteins, which is what lets it draw a network this size at all; with exact repulsion it is a second’s work only to about twelve hundred. At this size the force layout says how dense the network is and little else, which is what the modules view (§8) is for.

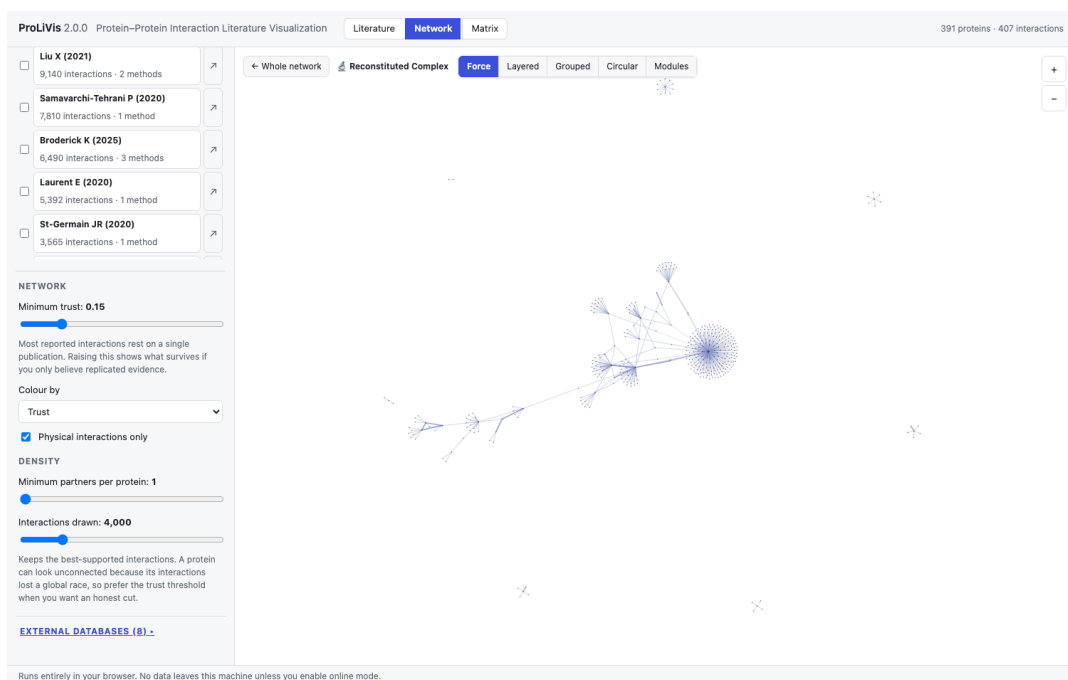


Figure 11: A sparse network: everything reported by Reconstituted Complex, which falls into many components. Each is laid out on its own and they are packed around the largest, spread evenly. Left to one embedding, a component with no edge to the rest feels only repulsion and drifts far outside the others, so fitting the view to include it shrinks the part worth looking at.

7 One protein, and the evidence behind it

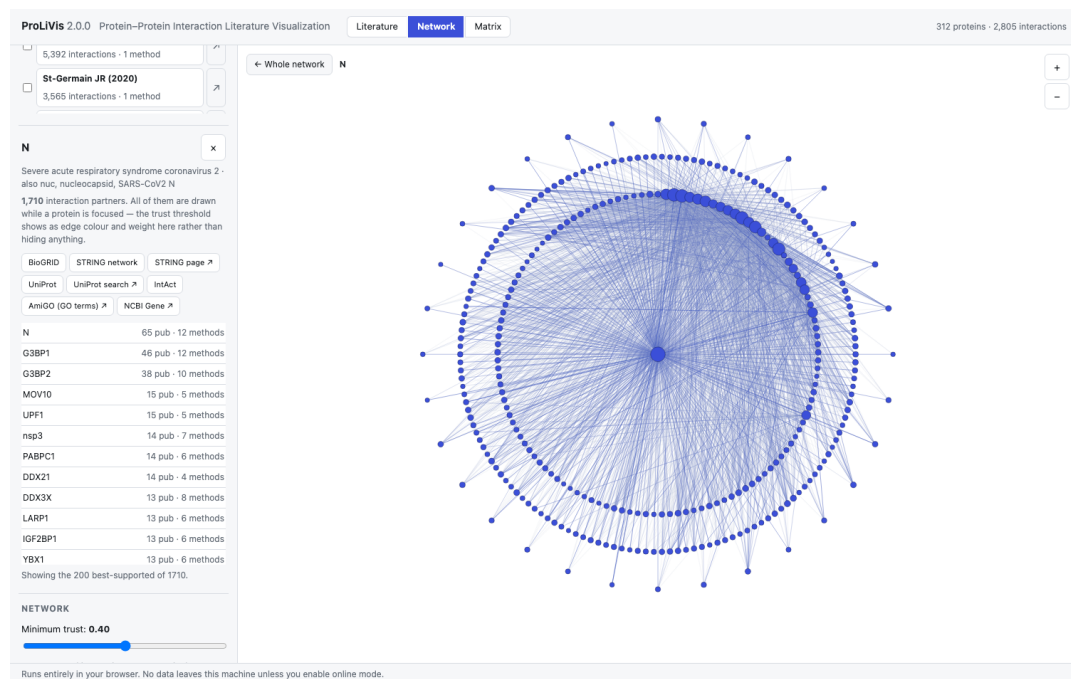


Figure 12: Clicking a protein centres the view on it and lists every interaction it takes part in, with how many publications and how many methods support each. The arrangement becomes an ego layout: radius is graph distance, so the picture states how each partner is reached. The list says how many of the partners it is showing.

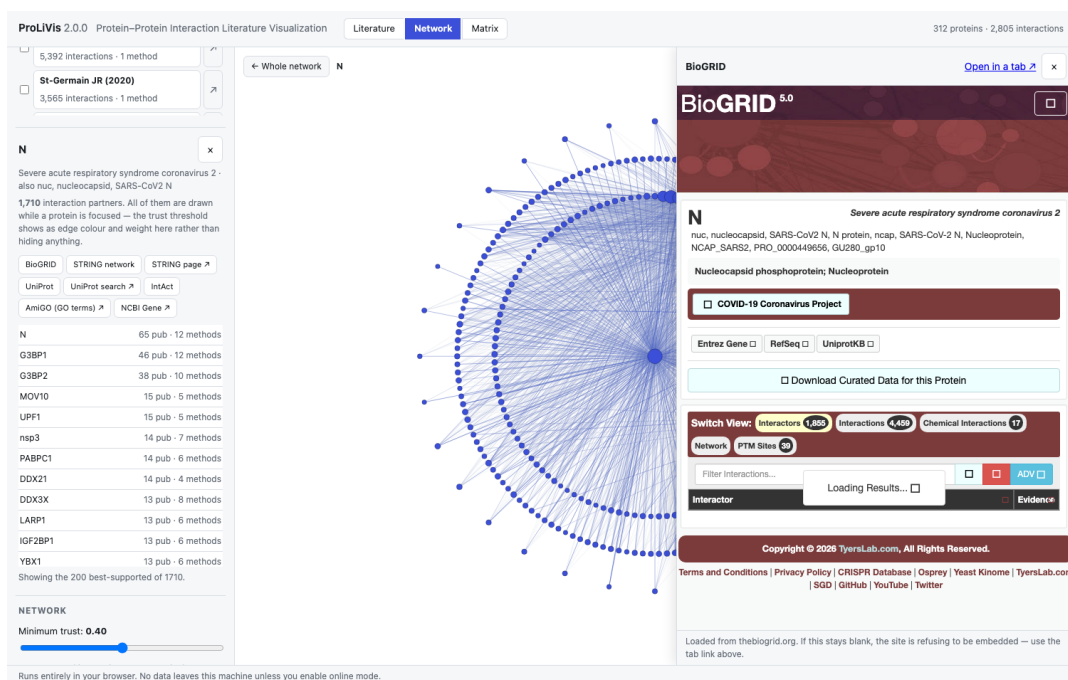


Figure 13: Links out to BioGRID, STRING, UniProt, IntAct, AmiGO and NCBI Gene. Where a site permits framing its page opens beside the network; where it does not, the link opens a tab, and STRING's network image is shown instead. The list is editable: a base URL and a template over {symbol}, {biogridId}, {entrez}, {swissprot} and {organismId}. Nothing is contacted until a link is clicked.

8 Reading a network one level up

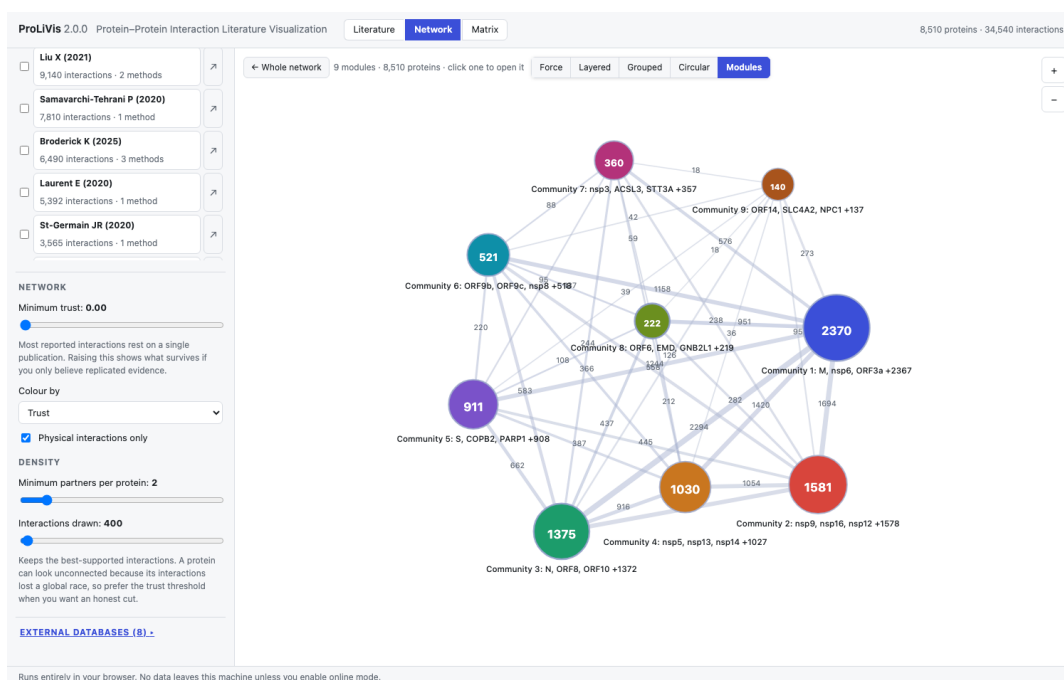


Figure 14: Modules. A node is a set of proteins — area by membership, the count printed inside, the ring the mean trust of its internal interactions — and a link is the evidence spanning two sets. Biconnected components where they decompose the network, trust-weighted communities where they do not.

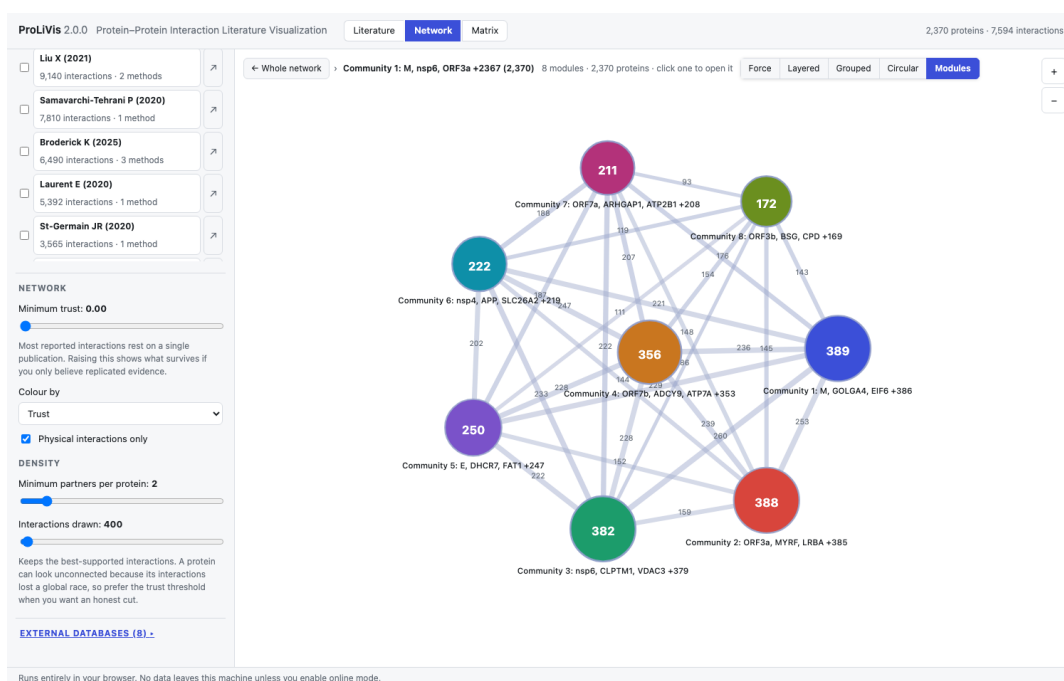


Figure 15: One module opened. If what is inside is still too large to draw it is contracted again, and the breadcrumb grows a level; any level in the trail returns to it. On the full human interactome this is four steps from a million interactions to 53 proteins.

9 The adjacency matrix

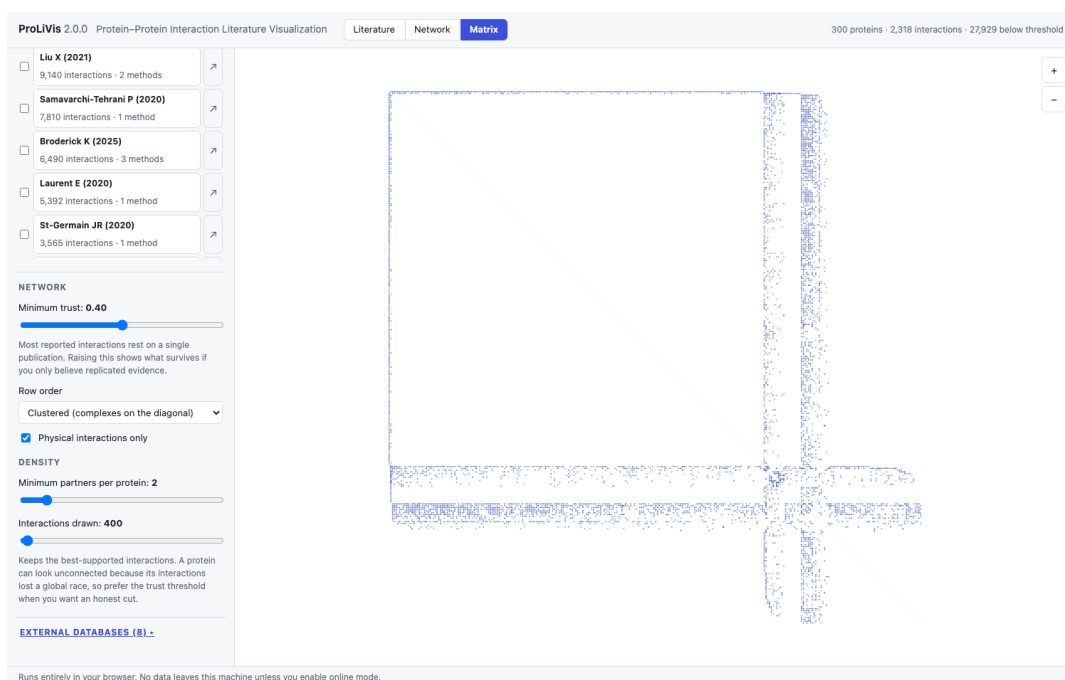


Figure 16: The same interactions as a reorderable matrix, coloured by trust. Rows are ordered by a breadth-first traversal within components, largest first, visiting the highest-degree neighbour first, which puts each module in a contiguous block — so complexes appear as blocks on the diagonal. The matrix is symmetric and both triangles are drawn: a triangular plot is more compact and far harder to read one protein's row from.

10 Searching the literature, and keeping some of it

☐ 5,392 interactions · 1 method
 ☐ **St-Germain JR (2020)**
3,565 interactions · 1 method
 ☐ **Gordon DE (2020)**
1,028 interactions · 4 methods
 ☐ **Chen Z (2021)**
437 interactions · 3 methods

[EXTERNAL DATABASES \(8\) ▾](#)

FILTER PUBLICATIONS

Drawing **1,688** of 1,688 publications. Counted within this organism and the methods selected — the same quantity the node sizes encode, so a paper's total across the whole release can be larger.

Interactions per publication

to

1–7,810 in this literature.

Proteins per publication

to

1–2,757 in this literature.

Methods all 19

☒ Reconstituted Complex 1,082
☒ Co-crystal Structure 477
☒ Affinity Capture-Western 307
☒ Co-localization 95
☒ Biochemical Activity 91
☒ Affinity Capture-MS 78
☒ Co-purification 49

LAYOUT

COLLECTION (3) [clear](#)

Samavarchi-Tehrani P (2020)
 7,810 interactions

Laurent E (2020)
 5,392 interactions

St-Germain JR (2020)
 3,565 interactions

Up to 16,767 interactions before overlaps are merged — an interaction reported by two of these papers is one interaction with two pieces of evidence.

Visualize combined network

Save as a dataset

Figure 17: Search by author, year, PubMed id, DOI or enriched title; with an empty box the largest contributors are listed, which answers “what is this network mostly made of”. Each hit says how many interactions and how many methods it contributed before you commit to opening it. Ticking hits builds a collection that survives further searches.

☐ 5,392 interactions · 1 method

☐ **St-Germain JR (2020)**
3,565 interactions · 1 method

[EXTERNAL DATABASES \(8\) ▾](#)

COMPARE & MERGE

Compare with... ▾

FILTER PUBLICATIONS

Drawing 3 of 3 publications. Counted within this organism and the methods selected — the same quantity the node sizes encode, so a paper's total across the whole release can be larger.

Interactions per publication

to

3,565–7,810 in this literature.

Proteins per publication

to

1,023–2,269 in this literature.

Methods all 1

☒ Proximity Label-MS 3

LAYOUT

Fold methods with fewer than publications

Zero keeps every method separate.

Publications using several methods

Placed once, between their methods ▾

☒ Label publications when sparse

Export figure as SVG

☐ 5,392 interactions · 1 method

☐ **St-Germain JR (2020)**
3,565 interactions · 1 method

[EXTERNAL DATABASES \(8\) ▾](#)

COMPARE & MERGE

CORONAVIRUS 5.0.260 ▾

Only in Reading list	0
Only in CORONAVIRUS 5.0.260	39,740
In both	13,470
Agreement	25.3%
Shared, new evidence	4,704

Compared by BioGRID gene id. Merging keeps one copy of records present in both, so replication counts are not inflated.

Merge into a new dataset

FILTER PUBLICATIONS

Drawing 3 of 3 publications. Counted within this organism and the methods selected — the same quantity the node sizes encode, so a paper's total across the whole release can be larger.

Interactions per publication

to

3,565–7,810 in this literature.

Proteins per publication

to

1,023–2,269 in this literature.

Methods all 1

☒ Proximity Label-MS 3

Figure 18: A collection saved as a dataset of its own — scorable, comparable, exportable, and remembering which publications produced it — and then compared with the release it came from. Records are copied rather than referenced, so removing the source cannot change a figure built from the derived set, and overlapping evidence is merged rather than counted twice.

11 Views computed but not drawn

Four view models are implemented, tested and reachable from the API, but the interface does not yet draw them. They are listed here with their real output rather than described, and the honest summary is that they are data, not pictures.

Table 2: **upset**: which combinations of methods actually support an interaction. Most interactions rest on exactly one method; the largest intersection is the two mass-spectrometry techniques agreeing.

Method combination	Interactions
Proximity Label-MS	15,456
Affinity Capture-MS	8,124
Affinity Capture-MS + Proximity Label-MS	3,779
Co-fractionation	1,684
Protein-RNA	1,013
Affinity Capture-RNA	512
Protein-peptide	480
PCA	378

Table 3: **timeline**: when the evidence arrived, by method, over 2004–2025. The two dominant methods peak in 2020 and 2021, which is what a pandemic looks like in a database.

Method	Interactions	Peak year
Proximity Label-MS	26,776	2020
Affinity Capture-MS	16,970	2021
Co-fractionation	2,027	2025
Protein-RNA	1,650	2023
Affinity Capture-RNA	1,614	2023
Reconstituted Complex	1,361	2021

Table 4: **methodChord**: which methods support the same interactions.

Method	Method	Shared interactions
Affinity Capture-MS	Proximity Label-MS	4,243
Affinity Capture-MS	Protein-RNA	365
Affinity Capture-MS	Co-fractionation	263
Affinity Capture-MS	Affinity Capture-Western	243
Protein-RNA	Proximity Label-MS	222
Affinity Capture-MS	Affinity Capture-RNA	221
Affinity Capture-MS	Two-hybrid	192
Co-fractionation	Proximity Label-MS	174

bipartite places publications and the proteins they touched on two axes with a lane per method: over the twelve largest publications here it produces 212 nodes, 1,511 links and 3 lanes, and reports that it truncated. Each of these takes an assembled input, and the API produces those inputs from a query (**pairSystems**, **timelineRecords**, **bipartiteInput**).

12 The trust model, and what can be done with it

Every interaction is scored from seven terms; the manuscript gives the equations. What matters in use:

- Terms whose inputs are unknown are reported as *unknown*, not as zero, and the remaining weights are renormalized. A **coverage** figure accompanies every score, saying how much of the model was informed.
- Three presets ship: **evidence-only** (fully offline), **literature-aware** (the default) and **structural-strict**.
- Re-weighting is instant, because evidence is gathered once and scoring is arithmetic over it — **gather** then **rescore**.
- **calibrate** and **ablate** measure a configuration against a reference set of known interactions, reporting AUROC, average precision and a leave-one-term-out ablation. The manuscript does not report numbers from these because that needs an external reference set; the harness is here.

```
const gathered = await prolivis.gather({ datasetId, organismId: 2697049 })
const strict   = prolivis.rescore(gathered, 'structural-strict')
const custom   = prolivis.rescore(gathered, { weights: { independence: 0.4 } })
```

13 Structure, and what survives the evidence

All of these are API-level and all take the graph built at a chosen trust threshold, so every structural question can be asked of “the network I actually believe”.

```
const graph = await prolivis.graph({ datasetId }, { minScore: 0.4 })

prolivis.cliques(graph, { minSize: 3 }) // candidate complexes
prolivis.modules(graph)                // biconnected components,
                                        // articulation points, bridges
prolivis.cores(graph)                  // k-core decomposition
prolivis.communities(graph)            // Louvain, trust-weighted
prolivis.autoContract(graph)           // what the Modules view draws
prolivis.removeEdges(graph, 'trust-ascending')
```

Edge removal comes in two orders. **betweenness** is classical Girvan–Newman and asks where the joins are. **trust-ascending** removes the least-supported edge first and asks what survives if only the evidence is believed; both record component count and modularity at every step.

14 Export, and reproducing a figure

```
prolivis.exportTable(scored, 'csv') // a column per trust term
prolivis.exportSif(scored)          // Cytoscape
prolivis.exportGraphml(graph)       // typed attributes
prolivis.toSvg(scene, 'title')      // the picture, as vector

const manifest = prolivis.manifest({
  dataset, query, trust, layout, publicationFilter,
})
const text = prolivis.serializeManifest(manifest)
```

The manifest records the release, the query, the filters, the trust configuration and the layout options. With a deterministic layout that is sufficient to regenerate a figure exactly — which ProLiVis 1.0 could not do even for its own author, its layout engine having been an external binary that is now lost. `checkManifest` reports whether the loaded datasets can reproduce a given manifest, naming what differs.

15 Known gaps

Stated rather than left to be discovered:

- Online mode against the BioGRID REST service exists in the API but has no panel in the interface; an access key cannot yet be entered by hand.
- Data export (CSV, GraphML, GML, SIF) and the session manifest are API-only. The interface exports the figure as SVG.
- Trust weights are editable through the API but not in the interface, which offers the threshold slider and the shipped presets' behaviour only.
- UpSet, timeline, chord and bipartite are computed but not drawn (§11).
- Literature enrichment covers OpenAlex with PubMed as a fallback; publications neither resolves leave the two bibliometric terms unknown, which the coverage figure reports.